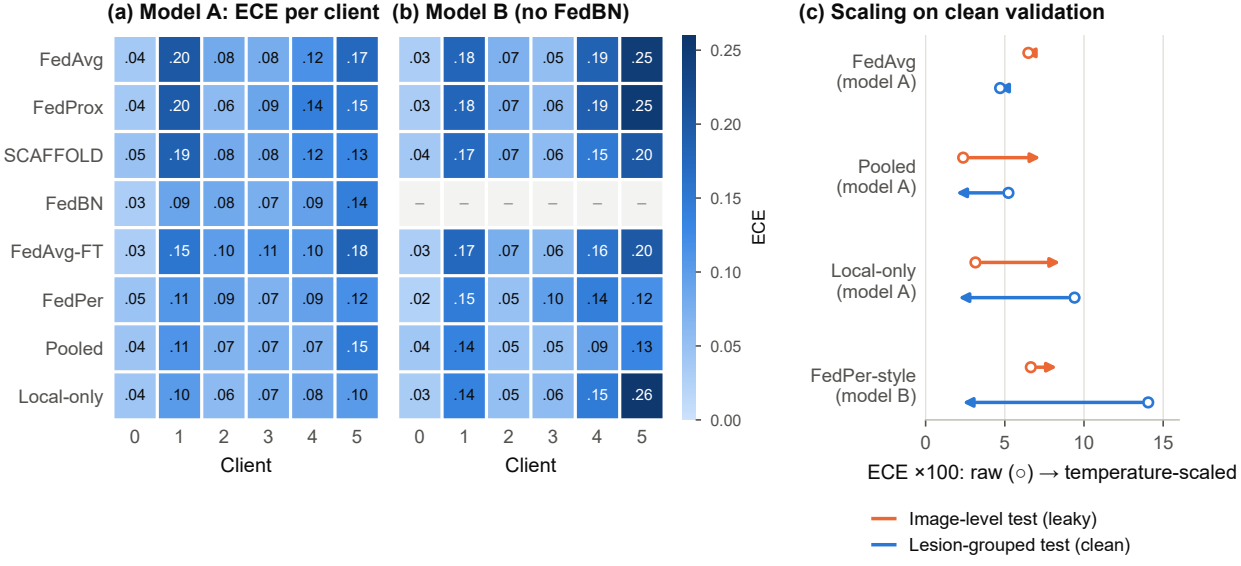

(a) Model A: ECE per client
(b) Model B (no FedBN)
(c) Scaling on clean validation
FedAvg
FedProx
SCAFFOLD
FedBN
FedAvg-FT
FedPer
Pooled
Local-only
.04 .20 .08 .08 .12 .17
.04 .20 .06 .09 .14 .15
.05 .19 .08 .08 .12 .13
.03 .09 .08 .07 .09 .14
.03 .15 .10 .11 .10 .18
.05 .11 .09 .07 .09 .12
.04 .11 .07 .07 .07 .15
.04 .10 .06 .07 .08 .10
.03 .18 .07 .05 .19 .25
.03 .18 .07 .06 .19 .25
.04 .17 .07 .06 .15 .20
– – – – – –
.03 .17 .07 .06 .16 .20
.02 .15 .05 .10 .14 .12
.04 .14 .05 .05 .09 .13
.03 .14 .05 .06 .15 .26
0 1 2 3 4 5
Client
0 1 2 3 4 5
Client
ECE
0.25
0.20
0.15
0.10
0.05
0.00
FedAvg (model A)
Pooled (model A)
Local-only (model A)
FedPer-style (model B)
0 5 10 15
ECE ×100: raw (○) → temperature-scaled
Image-level test (leaky)
Lesion-grouped test (clean)